

DataLink and Product Streamer

10 GB Download and SHA-256 Test

Download start: 05 October 2026, 08:25:19 UTC

Endpoint: http://64.176.188.89:18080

Product: lab.test:large-1tb.fits

Test configuration

A 10-second, single-stream reverse TCP iperf3 baseline was followed by a 10 GB download from the temporary VM to the client. DataLink resolved the synthetic product; Product Streamer delivered bytes 0-9999999999 from the approximately 1 TB sparse FITS source. The download had no total time limit. Blocks were hashed, checked and discarded. A reference SHA-256 was independently computed from the same byte range on the server before the network tests.

Results

Measurement	Result
iperf3 receiver throughput	137.746 Mbps / 16.421 MiB/s
iperf3 duration / received bytes	10.003 s / 172,228,608 bytes
Requested / received download bytes	10,000,000,000 / 10,000,000,000
Download duration	576.548 s (9.61 min)
Download average throughput	138.757 Mbps / 16.541 MiB/s
Time to first byte	1.287 s
Download / iperf3 throughput	100.73%
Transfer completion	PASS
SHA-256 comparison	PASS
Synthetic header / zero-data check	PASS

Data integrity verification

The requested byte count and HTTP range were verified. The received SHA-256 matched the independently computed source-range digest; FITS header markers and zero-filled payload checks also passed. Verification covers the complete downloaded range.

Received SHA-256:

84b841e5ef821e36864a1c02573d509d879aeea3d9c32ed06b69faddc391e032

Reference SHA-256:

84b841e5ef821e36864a1c02573d509d879aeea3d9c32ed06b69faddc391e032

Evidence: lab/evidence/large-speed/20261005T082429Z/speed.json